

Stage 1 Tutorial Activity

- Why Software Engineering Still Matters

Stage 1 | Introducing Software Technology Case Study with Python and Guided AI use

Learning goals

- Explain why software engineering is broader than coding.
- Identify stakeholders in a simple software problem.
- Recognise missing requirements.
- Critically evaluate AI-generated feature suggestions.
- Explain why AI output should not automatically be treated as correct.

Activity 1 - Think-Pair-Share (10 minutes)

If ChatGPT or Copilot can produce a 100-line Python application very quickly, what knowledge does a software engineer still need?

1. Problem solving skills/knowledge because still being able to understand what needs to be built as well as why it needs to be built.
2. System design and knowing how different components, databases, APIs, and other services/systems can connect or fit together.
3. Code review/debugging is important knowledge because checking AI generated code for bugs, security issues, and for performance issues.

Activity 2 - Is This Software Engineering? (10 minutes)

Scenario A: A student writes a 50-line Python calculator.

Scenario B: A team develops a payroll system used by 5,000 employees.

Scenario C: An AI assistant generates a simple appointment application from one prompt.

Scenario	Programming?	Software engineering?	Why?
A	Yes	No	It involves writing code to solve a simple problem, but doesn't include any software engineering practices like scalability, ongoing maintenance and any team collaboration.
B	Yes	Yes	Involves writing code while also involving with some software engineering practices like team collaboration,

			system security, possibly some testing/troubleshooting, scalability, and possibly some long-term maintenance for users.
C	No	No	The user is using generative AI to produce code for them, rather than programming it themselves, and simple, vague prompts of generation usually doesn't include strong system structure, poor validation, no room for future maintenance.

Activity 3 - SmartCare Problem Analysis (20 minutes)

Client statement: SmartCare Community Clinic currently uses spreadsheets and paper records to manage patients and appointments. The clinic wants new software to improve these processes.

Task 1 - Identify stakeholders

Stakeholder	What do they need?
Customer/Patients	Smooth and fast booking experience, easy access to medical records
Receptionists	Straight forward easy to understand system to schedule appointments with almost to no risks of any errors like double bookings etc.
Doctors/Medical Staff	They need to have quick and reliable accessibility to complete patient medical histories during any consultations.
Clinic Owners	They need efficient and smooth operating workflow, data security compliance, and reporting tools to keep track of financials, operating efficiency etc.

Task 2 - Identify current problems

1. Paper records can easily be physically lost, damaged, or misfiled, which creates a high risk of permanent data loss.
2. Multiple staff members updating shared spreadsheets often leads to duplicate entries, typing errors and possibly some version control issues.
3. Paper documents and unencrypted spreadsheets lack strong access controls, which creates serious security risks and possibly putting the patient's privacy at risk.
4. Searching through physical files and manually updating spreadsheets slows down daily workflow and operations, resulting in longer wait times for patients.

Task 3 - Ask client questions

1. What specific patient details and medical history need to be stored in the new system?
2. Who will need access to the software, and what permission levels should different staff roles have?
3. What are your budgeting limits and expected timeline for deployment/release?
4. Are there specific privacy laws or security conditions that the software must comply with?
5. Do you need the new software to integrate with any existing systems?

Activity 4 - Critique an AI Response (15 minutes)

An AI assistant suggests: appointment management; facial-recognition login; AI diagnosis recommendations; patient search; online payment; practitioner schedule view; insurance processing; automatic treatment-plan generation.

Suggestion	Client evidence?	In scope?	Decision
Appointment management	Yes, client mentions managing appointments	Yes	Include
Facial recognition login	No, client never asked for complex authentication methods	No	Exclude
AI diagnosis recommendations	No, client requested operational improvements, not automated clinical diagnosis	No	Exclude
Patient search	Yes, client needs to manage patients currently held in spreadsheets/physical files	Yes	Include
Online payment	No, client statement focuses on managing patients and appointments	No	Exclude
Practitioner schedule view	Yes, managing and viewing appoints efficiently	Yes	Include
Insurance processing	No, client did not mention handling insurance claims or billing	No	Exclude
Treatment-plan generation	No, client requested management tools, not automated medical treatment generation	No	Exclude

Exit question

Write one activity that a software engineer must perform and that cannot safely be delegated entirely to AI.

- An AI model cannot talk to clinic staff to figure out their real world disruptions, handle office relationships, or take legal responsibility for project decisions.